

Assessing Ranking Fidelity of Competition Structures via Weighted Mutual Information

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Abstract

We evaluate multiple competitions structures to assess their effectiveness in accurately reproducing the true underlying ranking of participating teams. Using repeated simulations for each structure, we compare teams' true performance ranks to their simulated outcomes. To quantify the accuracy of each format, we calculate

weighted mutual information between true and simulated rankings across a large number of replicates. “Salient” weights are applied to look at only the top i out of n total teams/competitors in a given competition structure. This information-theoretic approach allows us to objectively compare formats and identify which structures most effectively preserve rank order. Results are visualized through comparative plots, providing clear insights into the trade-offs and strengths of each competition design.

Keywords: tournament structures, mutual information

1 Introduction

There are numerous different competition structures that dictate how a set of teams compete against each other to determine an overall ranking. One of the most common structures is the bracket tournament where if a team wins their competition, they advance to the next round. Bracket tournaments can be run as single elimination where competitors advance or are eliminated based on a single match (e.g. NCAA College Basketball Tournament, National Football League (NFL) playoffs, major tennis tournaments) or where each round is contested as a best of k series (e.g. $k = 7$: National Basketball Association (NBA) playoffs, National Hockey League (NHL) playoffs, k varies by round: Major League Baseball (MLB), Women’s National Basketball Association (WNBA)) and the series winner advances while the series loser is eliminated. As early as Carroll (1883), single-elimination bracket tournaments were questioned for their lack of reliability, noting that strong players may be eliminated early due to unfavorable pairings or random variation, and proposed reforms to reduce the role of luck in determining outcomes.

Additional styles of the bracket tournaments include variations where competitors are not eliminated after a single loss such as double elimination tournaments where competitors continue until they have lost twice. Within the double elimination tournament, there are several variations. For instance, the NCAA College Wrestling tournament is a double elimination tournament, however, if a competitor loses a match they fall into the consolation bracket and the best they can finish after a loss before the finals is 3rd place making this not a “true” double elimination format. Olympic judo uses a variant of this system with single elimination until the quarterfinals followed by a consolation bracket consisting of the four quarterfinal losers who still have a chance for a bronze medal¹. Aziz et al. (2017) show that, even within a fixed bracket format, the arrangement of the draw can significantly affect outcomes: determining whether a specific player can be guaranteed a win by optimally arranging the bracket is an NP-complete problem.

¹in Olympic judo they award two bronze medals to the two competitors who each win their side of the consolation bracket

“True” double elimination - a bracket tournament where a competitor can still finish ranked first as long as a competitor has not lost two games - is used in NCAA baseball and softball for certain rounds in their postseason. An additional variant of double elimination include double elimination brackets with a “true third” match which includes an additional match contested between the competitors ranked 2nd (loser of the finals) and 3rd (winner of the consolation bracket) at the end of the tournament *if they have not already competed against each other*. There is also a variation of double elimination where only those competitors who have lost to a competitor that makes the finals are placed into the consolation bracket and given the chance to compete for third place. This style of competition is used in Olympic wrestling (both freestyle and Greco-Roman)².

One alternative to bracket style tournaments are adaptive pairing systems such as the Swiss system, in which competitors are paired each round based on their current record, so that similarly performing players face one another without any competitor being eliminated. Glickman and Jensen (2005) extended this idea by proposing a Bayesian adaptive pairing method that incorporates prior information about participant strength, determining pairings by maximizing the expected information gained about the relative strengths of competitors. Kobayashi et al. (2006) further address scheduling efficiency by proposing an algorithm that uses parallel pairings, where multiple matches occur simultaneously, to reduce the total number of rounds required to establish a full ranking, drawing on lattice path counting techniques from ballot problems to determine the minimum number of rounds needed.

In bracket style competitions, only a handful of the match-ups that are possible actually occur. When every possible competitive pairing occurs, this is referred to as a round-robin structure where every competitor plays every other competitor the same number of times. Scheid (1960) formalized the structure of round-robin tournaments, deriving general equations for the number of games required for any number of players and teams, including cases where each player competes with every other player exactly once or exactly twice. Single round-robin, where each possible match-up occurs exactly once, is used in, for example, the group stage of the FIFA World Cup to create a ranking to determine which teams advance to the next round. Double round-robin, where each possible pairing occurs twice, is used in association football in English Premier League (EPL) and in chess during the FIDE candidates tournament (a tournament that decides who will challenge the world champion). Suksompong (2016) further studies the scheduling problem within round-robin formats, examining asynchronous tournaments where no two games are played at the same time and proposing schedules that optimize rest time, parity in games played, and fairness across competitors. For the sake of brevity, we review only a small number of possible competition structures, but the interested reader can find a more complete list of competition structures

²As in Olympic judo, Olympic wrestling also award two bronze medals to the winners of each side of the consolation bracket

in K. Devriesere et al. (2025), Appleton (1995), Csató (2021), and Karel Devriesere et al. (2025), which offers a broad comparative overview of tournament formats evaluated along the dimensions of efficacy, fairness, attractiveness, and strategy-proofness.

A natural question of interest when considering different types of competition structures is: which competition structure is best? As with most complicated questions, the answer is “it depends.” One commonsense conceptualization for the best competition structure is that the final rankings produced by a competition structure should closely match the true rankings of the strength of the competitors. That is, a good competition structure should rank the true best team first, the true second best team second, and so on, until finally ranking the true worst team last. We refer to this property of a competition structure as the efficacy of the structure (Lasek and Gagolewski (2018), Sziklai et al. (2022)). Lasek and Gagolewski (2018) finds through simulation that the most efficacious league format of the systems they considered involves a two-stage system with a three-round-robin followed by a single round-robin, though such formats require a large number of matches. Sziklai et al. (2022) similarly find through simulation that the Swiss system is among the most accurate formats, particularly because of its ability to rank all participants, and that seeding can improve the efficacy of knockout and group-based designs, though this effect is marginal unless one has an unusually accurate prior estimate of the true ranking.

Rather than being interested in a full ranking of competitors, a tournament organizer may only be interested in correctly ranking the true best competitor as first, with the ranks of the remaining competitors viewed as inconsequential. We refer to this measure of a tournament as effectivity (Glenn (1960)). Glenn (1960) provided one of the earliest formal comparisons of tournament formats, evaluating six designs for four-player competitions, including round-robin, single-elimination variants, and double-elimination. They found that tournaments with more games are more accurate but less efficient, with round-robin formats maximizing ranking accuracy.

While there are many competition structures where efficacy and effectivity are useful measures³, there are many other competitions where the correct ranking of more than one competitor, but not all the competitors, is of import. For instance, in Olympic competition, the goal of a competition structure is to correctly rank the top 3 competitors who are awarded the gold, silver, and bronze medals, respectively⁴. In many American sports, for instance the NFL, the goal of the competition structure of the regular season is to rank the top 7 teams in each conference to choose the playoff teams with the ranking of the teams 8 through 16 in each conference inconsequential in determining a Super Bowl champion⁵. We coin the term

³efficacy is a useful measure for the EPL, for example, as the ranking of the teams matters at both the high end (i.e. entry into special tournaments) and the low end (i.e. relegation status) and effectivity is a useful measure for the NCAA College Basketball tournament (i.e. main goal is to crown a champion)

⁴Some sports award two bronze medals (e.g. wrestling, judo, boxing, tae kwon do)

⁵The ranking of teams outside of the playoffs does matter for draft purposes though

“salient ranks” to refer to the ranks of a competition structure that matter to the organizer. That is, the ranks 1, 2, and 3 in Olympic competition are the “salient ranks” in that competition structure. In this notation, efficacy is a measure with salient ranks 1 through n where n is the number of competitors, and effectivity is a measure where 1 is the only salient rank.

A prerequisite for measuring how well a competition structure recovers true rankings is a reliable model of the probability that a stronger competitor defeats a weaker one in any individual contest. Mosteller (1952) addressed this directly using historical World Series data to estimate the probability that the better team wins a single game and demonstrating that even in a best-of-seven series the stronger team often does not prevail. This highlighted the fundamental role of chance in tournament outcomes and linked the number of games played to the probability that final results reflect true team quality. Bradley and Terry (1952) formalized this probabilistic framework by introducing a model that assigns a positive strength parameter to each competitor and defines the probability that one competitor beats another as a function of the difference of their strengths. This formulation has become a cornerstone of modern statistical ranking theory and is widely used in sports analytics and preference modeling. Glickman and Jones (2025) provides a contemporary overview of this broader class of statistical models, reviewing methods from Bradley-Terry and Thurstone-Mosteller (CITATION?) through modern adaptive systems such as Elo (CITATION) and Glicko (CITATION), with particular attention to models that allow competitor ability to change over time using Bayesian updating frameworks.

If the only goal of a competition structure was high efficacy of the salient ranks, the best way to achieve this would be to play each pair of possible match ups of competitors as many times as possible. Of course, this is impossible due to many factors including time constraints, cost constraints, and wear and tear on competitors in high impact sports (e.g. boxing, American football, etc.). In addition, competitions where the true best team is guaranteed or nearly guaranteed to win may be less appealing to fans and may drive down demand (Johnson and Fort (2022)). Scarf and Yusof (2011) demonstrate this tension empirically using simulation based on international soccer data, showing that seeding increases the probability that the best teams advance but at the cost of reducing overall outcome uncertainty, which is a dimension of tournament design that matters to fans and organizers alike. Krautmann et al. (2011) extends this to the concept of “playoff uncertainty”, defined as uncertainty about whether a team will qualify for the postseason at all, and, using 50 seasons of MLB attendance data, finds that tighter pennant races boost game attendance, particularly toward the end of the season.

In light of this, many different competition structures have been proposed, several of which are mentioned previously. Given these many competition structures, ideally there would be a way of comparing the performance of different competition structures to each other. A natural approach is to compare the rankings that different structures produce. Kendall

(1938) proposed τ , a measure of rank correlation that takes the value +1 when two rankings agree perfectly and -1 when they are exactly reversed, providing a principled way to compare any two orderings. Kendall and Smith (1940) further explored situations where such comparisons are complicated by inconsistencies: cases where preferences are “circular,” such as A preferred to B, B to C, and C to A. They developed measures of agreement among multiple judges that are grounded in Kendall’s τ . Slater (1961) addresses this problem of circularity directly, defining the minimum number of preference reversals needed to make an inconsistent set of paired comparisons transitive, thereby recovering a consistent linear ranking. Beyond full-ranking comparisons, Shieh (1998) proposes a weighted variant of Kendall’s τ that places greater emphasis on agreement at the top of the ranking, and Fagin et al. (2003) introduce distance measures specifically designed for comparing partial or “top k” lists. Measures like these are well suited to settings such as ours where only salient ranks matter.

Here, we propose a novel metric for evaluating different competition structures using an information theoretic approach. As an analogy, we view the results of a tournament as a transmitted message, and we use information theory to assess how accurately the message (i.e. the tournament results) matches the true transmission (i.e. the true ranking of competitors).

Specifically, we use weighted mutual information to compare between the ranks generated by a competition structure and the true ranks of the competitors. Additionally, as part of the weighting function, we include information to specify the salient ranks via an indicator vector allowing us to compare different competition structures with different choices for salient ranks.

The remainder of this manuscript presents the mathematical foundation of the proposed measure followed by a simulation study estimating our proposed measure for different types of competition structures with varying choices of salient ranks. We close by offering a summary of the work and ideas for future directions.

2 Methods

Consider a set of n teams indexed from $i = 1, 2, \dots, n$ each with an associated true strength parameter $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_n)$ such that $(\theta_1 \geq \theta_2 \geq \dots \geq \theta_n)$. Next, let $r(\boldsymbol{\theta})$ be the indexes of $\boldsymbol{\theta}$ when $\boldsymbol{\theta}$ is sorted from largest to smallest so that $r(\boldsymbol{\theta}) = \{1, 2, \dots, n-1, n\}$ (Note: While we allow for two values of $\boldsymbol{\theta}$ to be equal to each other, we choose to assign unique ranks to each θ_i). Next define $T(\phi, s)$ as a tournament with schedule ϕ and seeding structure s . ϕ contains all the information about the scheduling of teams, which could be fully known prior to the tournament (*e.g.* round robin) or determined as the tournament progresses (*e.g.*

single elimination tournament). We then let $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ be an n -dimensional vector-valued random variable that gives the results of a tournament as a vector of the indexes of $\boldsymbol{\theta}$. The number in the first position of the vector $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ indicates the index of the team that finished first, the number in the second position of the vector $\hat{r}(T(\phi, s), \boldsymbol{\theta})$ indicates the index of the team that finished second, and so on. (Note: We do allow for ties in $\hat{r}$.)

For example, in a $n = 4$ team tournament, $\hat{r}(T(\phi, s), \boldsymbol{\theta}) = (3, 2, 1, 4)$ indicates that the team with index 3 (i.e. the true third best team) won the tournament, the true second best team finished second, the true best team finished third and the true 4th team finished 4th in that particular tournament. If $\hat{r}(T(\phi, s), \boldsymbol{\theta}) = (1, 2, 3, 4)$, this means that the random outcome of the tournament was the same as the true ordering of the teams.

As noted above we allow for ties in $\hat{r}$. Continuing the above example, if $\hat{r}(T(\phi, s), \boldsymbol{\theta}) = (3, 2, 1, 3)$ this indicates that the teams that are truly ranked 1st and 4th tied for 3rd.

We need a transition: If one views the true ranking of the teams in a tournament as a message to be sent to a receiver and the outcome of the tournament as a message that is received, we can measure the “goodness” of a tournament in terms of its ability to accurately transmit the true ranking. We can then use concepts from information theory to assess the ability of a tournament to correctly rank teams. Specifically, we start with the concept of mutual information (Guinasu (1977)).

For two random variables X and Y mutual information is defined as:

$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$$

In our setting, we replace X and Y with $r(\boldsymbol{\theta})$ and $\hat{r}(T(\phi, s), \boldsymbol{\theta})$, and we could seek to estimate $I(r(\boldsymbol{\theta}), \hat{r}(T(\phi, s), \boldsymbol{\theta}))$. For simplicity, we drop the arguments from the functions r and $\hat{r}$ for ease of exposition.

In order to compute mutual information, marginal distributions of r and $\hat{r}$ are required. Without knowing r , all possible permutations of $\hat{r}$ are equally likely so we set $p(\hat{r}) = \frac{1}{n!}$. By a similar argument we set $p(r) = \frac{1}{n!}$.

We define the mutual information of r and $\hat{r}$ to be:

$$I(r, \hat{r}) = \sum_r \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})} = n! \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{(\frac{1}{n!})^2}$$

While there are $n!$ different permutations for the result of r , we don't need to sum across these as any specific choice of r is just an arbitrary labeling of the true strength parameters vector $\boldsymbol{\theta}$. So given r , the distribution of $\hat{r}$ is the same up to the labeling. Therefore, we

only need to consider a single permutation of r , we compute the quantity inside the summation, and then multiple by $n!$ (effectively summing across r).

In order to compute this quantity, we need to compute $p(r, \hat{r})$. This probability is found by calculating the probability of a given permutation of $\hat{r}$ and r is assumed to be the permutation from $\{1, 2, \dots, n\}$. We estimate $p(r, \hat{r})$ empirically through simulation.

However, using mutual information in this form does not suit our needs in this setting. The problem is that mutual information in this form will yield high values of mutual information when the output from the tournament $\hat{r}$ is highly consistent *even if the ranking from the tournament is incorrect* (i.e. the rankings produced by the tournament are very different from the true rankings). As an example, if $n = 4$ and the true order of θ is $r = \{1, 2, 3, 4\}$ and $p(\hat{r} = \{4, 3, 2, 1\}) = 1$ will be the same mutual information as when $r = \{1, 2, 3, 4\}$ and $p(\hat{r} = \{1, 2, 3, 4\}) = 1$ and in both of these cases the mutual information will be maximized at:

$$I(r, \hat{r}) = n! \sum_{\hat{r}} p(r, \hat{r}) \log \frac{p(r, \hat{r})}{\left(\frac{1}{n!}\right)^2} = n! \log((n!)^2)$$

and for the specific case when $n = 4$ would be:

$$4! \log_2(4!^2) = 24 * \log_2(24^2) = 220.0782$$

3 Weighted Mutual Information

In order to alleviate this problem, we instead consider *weighted* mutual information (Guisasu (1977)). We want to give more weight to permutations from $\hat{r}$ that are “closer” to r . Essentially, this problem comes down to choosing a metric that measures the closeness between the vectors $\hat{r}$ and r . One then gives more weight to combinations of $\hat{r}$ and r that are closer and less weight to combinations that are far apart. A natural choice for distance between vectors the sum of squared deviations between the two vectors. Specifically, we define the loss function between $\hat{r}$ and r to be $l(r, \hat{r}) = \sum_{i=1}^n (\hat{r}_i - r_i)^2$. Then, since small losses should be associated with large weights, we focus on the reciprocal of this loss function.

In general, any loss function $l(r, \hat{r})$ can be used to define the weighting function as follows:

$$w(r, \hat{r}) = \begin{cases} \frac{1}{l(r, \hat{r})}, & r \neq \hat{r} \\ 1, & r = \hat{r} \end{cases}$$

Note that in order to avoid dividing by 0, we define the weight to be 1 when r and $\hat{r}$ are exactly equal.

Specifically, when squared error loss is chosen, one gets the following weight function:

$$w(r, \hat{r}) = \begin{cases} \frac{1}{\sum_{i=1}^n (\hat{r}_i - r_i)^2}, & r \neq \hat{r} \\ 1, & r = \hat{r} \end{cases}$$

An alternative to defining weight this way is to consider looking at the correlation between $\hat{r}$ and r . Since, $\hat{r}$ and r are already ranks, the Pearson and Spearman correlation coefficients will be computationally exactly the same in this setting if that was the measure that someone chose to use.

Spearman correlation can be written as follows:

$$\rho_s = 1 - \frac{6 \sum_{i=1}^n (\hat{r}_i - r_i)^2}{n(n^2 - 1)}$$

,

which means that Spearman correlation is simply (one minus) a scaling of squared error loss. Since $\rho_s \in [-1, 1]$, but we desire weights $w(\hat{r}, r) \in [0, 1]$, we define $w(\hat{r}, r) = \frac{(\rho_s + 1)}{2}$ guaranteeing weights in the desired range.

While squared error loss and Spearman/Pearson correlation work well, in this setting we argue that Kendall's τ correlation is the most appropriate for weighting due to its interpretation and how this related to competitions. One way of interpreting Kendall's τ is that it's a scaling of the number of adjacent swaps required to go from one permutation of ranks to another. Recall that Kendall's τ is defined as:

$$\rho_\tau = 1 - \frac{2 \times \# \text{ discordant pairs}}{\binom{n}{2}}$$

. To convert ρ_τ back to the range between $[0, 1]$, which we want for weighting purposes, we subtract 1 and divide by 2 giving us $\frac{\rho_\tau + 1}{2}$. This then becomes:

$$w(\hat{r}, r) = \frac{\rho_\tau + 1}{2} = 1 - \frac{\# \text{ discordant pairs}}{\binom{n}{2}}$$

and the number of discordant pairs is equivalent to the number of adjacent swaps needed to get the order of $\hat{r}$ correct.

As an example, consider $r = (1, 2, 3, 4)$ and $\hat{r} = (1, 3, 4, 2)$. This requires two adjacent swaps to get $\hat{r}$ to equal r (i.e swap 4 with 2 and then 3 with 2). The Kendall's τ value between these two vectors is $\frac{1}{3}$. We would then define the weight between the two vectors to be $(\rho_\tau + 1)/2 = \frac{2}{3}$. This is exactly equivalent to $1 - \frac{\# \text{ adj swaps}}{\binom{n}{2}}$.

Once an appropriate weight function $w(r, \hat{r})$ is chosen, we get the following formula for weighted information:

$$I_w(r, \hat{r}) = \sum_r \sum_{\hat{r}} w(r, \hat{r}) p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})}$$

4 Tiered Ranking

This framework was developed to assess the efficacy of the full ranking of the top k teams generated by different competition formats. However, it may be of interest to have a competition structure that simply selects the top k teams in any order. Continuing with the Olympics example, if an organizer only cares about the top 3 teams earning medals *in any order* (i.e. $\hat{r} = \{1, 2, 3, 4\}$ and $\hat{r} = \{3, 2, 1, 4\}$ are equivalent), our framework can easily handle this modifying $\hat{r}$ (and r) so that the top k elements are ordered correctly and tied. That is, for the Olympic example, if k was 3 and $\hat{r} = \{3, 2, 1, 4\}$, we convert all the elements of $\hat{r} \leq k$ to 1 (i.e. $\hat{r}$ becomes $\{1, 1, 1, 4\}$), and then we proceed with the exact same formula. We refer to this as tiered ranking.

This concept can be extended to multiple tiers, which is applicable to a competition structure such as the premier league where the top 4 teams qualify for the Champions league and the bottom 3 teams are relegated. While this is straightforward to compute, we do not evaluate scenarios such as this in this article.

5 Salient Weights (and ranks)

In addition to the the weighting function w , we defined a second set of weights that we refer to as the “salient weights” and denote as the vector m . This set of weights is used to indicate which ranks have been determined to salient (as previously described) and consists of a vector of length n with elements that are positive for each of the salient ranks and 0 otherwise. As an example, for a four-team tournament, if the organizer decided that only the correctly choosing the best team as the winner was important, then $m = \{1, 0, 0, 0\}$ (i.e. measuring effectivity). In an event such as the Olympics, the salient weights could be 1’s for the first three elements and 0’s elsewhere such that $m = \{1, 1, 1, 0\}$. Additionally, one can also weight the salient ranks. For example, continuing the Olympics example where getting the top three correct is the main objective, one could argue that it is more important to get the first place correct than the third place. This could lead to, for instance, a salient weight vector of $m = \{5, 3, 1, 0, \dots, 0\}$ giving 5 times the weight to getting the first place team correct as getting the third team correct.

With the addition of the salient weighting vector, m , the formula for weighted mutual information becomes:

$$I_w(r, \hat{r}, m) = \sum_r \sum_{\hat{r}} w(r, \hat{r}, m) p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})}$$

Incorporating m into this function is straightforward for many weighting functions. We use the weighted Kendall's τ presented in Shieh (1998) with a slight modification to the sign function. Specifically, in the special case that $\text{sign}(0) = 1$. Then $w(r, \hat{r}, m)$ becomes:

$$w(r, \hat{r}, m) = \frac{\sum_{i \neq j} m_{ij}^* \text{sign}_{(0+)}(r_i - r_j) \text{sign}_{(0+)}(\hat{r}_i - \hat{r}_j)}{\sum_{i,j} m_{i,j}^* - \sum_i m_{i,i}^*}$$

where $m_{i,j}^* = m_{\min(i,j)}$ and $\text{sign}_{(0+)}(\cdot)$ is the sign of the input with the special case that a 0 maps to 1.

Finally, because this mutual information is unitless, the formula can be standardized to be between 0 and 1 as follows:

$$I_w^*(r, \hat{r}, m) = \frac{I_w(r, \hat{r}, m)}{H(r, \hat{r})}$$

where $H(\cdot)$ is the usual Shannon entropy. Using this framework, values of $I_w^*(r, \hat{r}, m)$ near 1 indicated competition structures with high efficacy. Values near 0.25 indicate tournament structures where all possible rankings are equally likely and values near 0 indicate rankings that are inefficacious (e.g. complete rank reversal).

Under the assumption of equal probability for all possible permutations of $\hat{r}$, we can simplify $p(r, \hat{r}) = \frac{1}{n!}$ and $\sum_r \sum_{\hat{r}} w(r, \hat{r}, m) = 0.5 \times n!$. Assuming that $p(r) = p(\hat{r}) = \frac{1}{n!}$, this results in:

$$I_w(r, \hat{r}, m) = \sum_r \sum_{\hat{r}} w(r, \hat{r}, m) p(r, \hat{r}) \log \frac{p(r, \hat{r})}{p(r)p(\hat{r})} = 0.5 \times n! \times \frac{1}{n!} \times \log\left(\frac{1/n!}{(1/n!)^2}\right) = 0.5 \times \log(n!)$$

Additionally, we can assume that r and $\hat{r}$ are independent and have equivalent probabilities of $\frac{1}{n!}$. Through the chain rule detailed in Cover and Thomas (2006),

$$H(r, \hat{r}) = H(r) + H(\hat{r}|r)$$

. Assuming independence and equality,

$$H(\hat{r}|r) = H(\hat{r}) = H(r)$$

. Thus,

$$H(r, \hat{r}) = 2 \times H(r) = -2 \sum_r p(r) \log(p(r)) = -2(n!) \times \left(\frac{1}{n!}\right) \times \log\left(\frac{1}{n!}\right) = \log((n!)^2)$$

and the final simplification:

$$I_w^*(r, \hat{r}, m) = \frac{I_w(r, \hat{r}, m)}{H(r, \hat{r})} = \frac{0.5 \times \log(n!)}{2 \times \log(n!)} = \frac{0.5}{2} = 0.25$$

6 Results

We evaluated tournament structures with $n = 16$ competitors to demonstrate the utility of our proposed weighted mutual information metric. Game outcomes were simulated probabilistically using a logistic model where the probability that competitor j defeats competitor k is given by:

$$p_{jk} = \frac{e^{\theta_j - \theta_k}}{1 + e^{\theta_j - \theta_k}}$$

where p_{jk} is the probability that competitor j beats competitor k and θ_j and θ_k are the strength parameters of teams j and k , respectively.

We focus on several simple competition structures focused mainly on variations of bracket tournaments to demonstrate the proof of concept of our proposed metric. Each competition structure was simulated 10,000 times to estimate $wI_m(r, \hat{r})$. In all bracket structures, seeding was assumed to be optimal, reflecting each team's true rank. For teams eliminated before the finals, ranks were assigned randomly within their possible range (e.g., semifinal losers were assigned rank 3 or 4 with equal probability).

We examined the following structures with the assumption of the ideal seeding for each competition:

- Single Elimination, seeding (1 vs 16, 8 vs 9, 4 vs 13, 5 vs 12, 3 vs 14, 6 vs 11, 7 vs 10, 2 vs 15)
- Double Elimination, seeding (1 vs 16, 8 vs 9, 4 vs 13, 5 vs 12, 3 vs 14, 6 vs 11, 7 vs 10, 2 vs 15)
- Round Robin
- Group Stage, 4 groups of 4, seeding (pot 1: 1-4, pot 2: 5-8, pot 3: 9-12, and pot 4: 13-16)

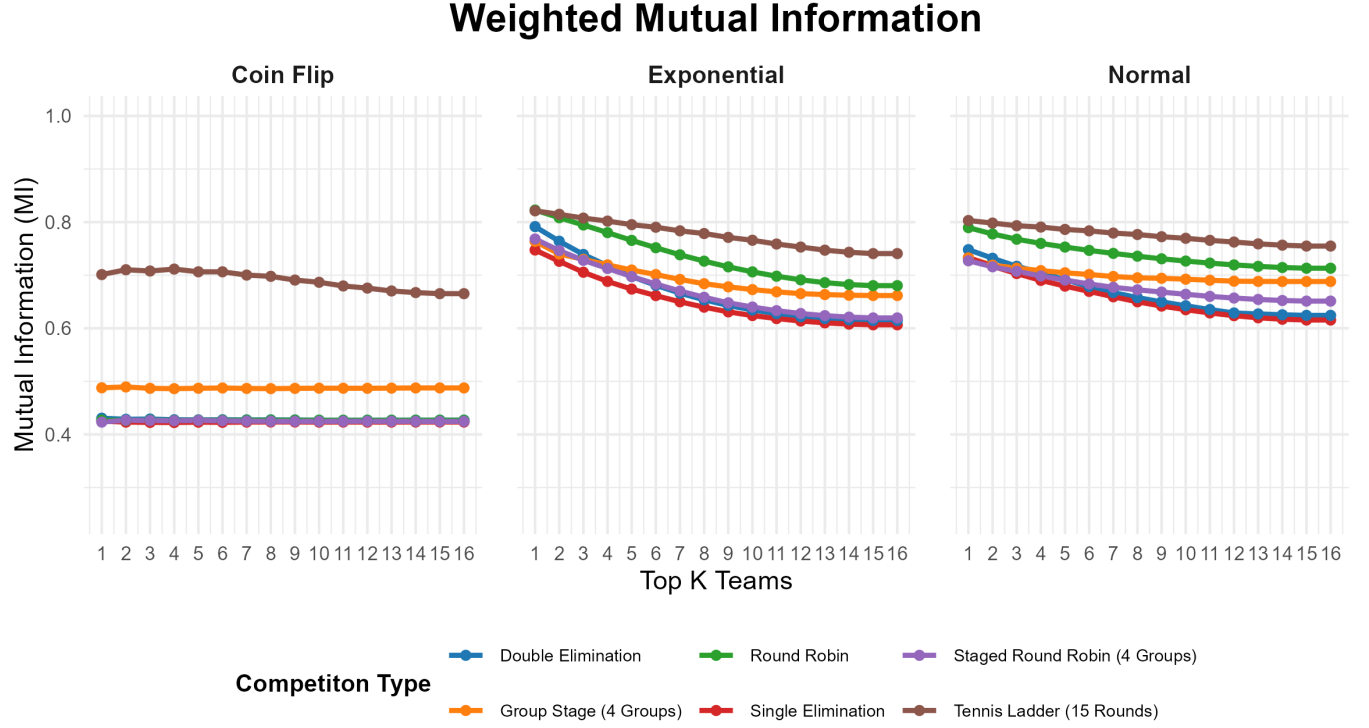


Figure 1: Weighted mutual information of various competition structures as a function of the top K teams for three distributions of strengths (Coin Flip, Exponential, Normal).

- Staged Round Robin, 4 groups of 4, seeding (pot 1: 1-4, pot 2: 5-8, pot 3: 9-12, and pot 4: 13-16)
- Tennis Ladder, seeding (16 vs 15, 16 vs 15 winner vs 14, etc.)

Figure 1 presents the normalized weighted mutual information $wI_m(r, \hat{r})$ as a function of the number of salient ranks (k) across multiple strength distributions. The tennis ladder structure consistently achieves the highest wI_m values across nearly all conditions, outperforming the traditional bracket formats regardless of k or strength distribution. This superior performance arises from the ladder’s inherent sequential constraint, which limits the extent to which highly-ranked teams can fall in the final standings $\hat{r}$. The structure essentially provides a built-in protection mechanism for top competitors while maintaining competitive integrity. However, this built-in protection is only applicable for the assumption that the seeding was done correctly.

Round robin tournaments achieve the highest wI_m value for $k = 1$ only under the exponential distribution, where the increased probability of top-seed victories (due to larger strength differentials) combines with the structure’s maximal number of matchups to provide the exception to the tennis ladder structure’s superior performance. However, this advantage

disappears under more balanced strength distributions.

6.1 Influence of Strength Distributions

The distribution of the strength parameters, θ , can substantially influence the competition’s information. Under the “Coin Flip” distribution, where every team has an equal 50% win probability, all structures exhibit reduced wI_m values compared to scenarios with heterogeneous strengths. This result is intuitive: when true skill differences are absent, tournament structures cannot extract meaningful ranking information beyond random assignment.

The exponential distribution, characterized by larger gaps between top competitors and smaller gaps among lower-ranked teams, produces distinct patterns. The disparity between single and double elimination widens considerably for small k , as the heightened probability of top-seed victories makes the recovery mechanism of double elimination particularly valuable for preserving information about elite competitors. Conversely, under normal and uniform distributions, with more evenly-spaced strength parameters, this advantage diminishes.

It is important to note that while the theoretical value for round robin for the distribution “Coin Flip” is 0.25, our simulations produced slightly different estimates due to finite simulation counts and inherent stochastic variation.

6.2 Tiered Ranking Analysis

Figure 2 examines the mutual information for tiered rankings of the top K teams. As K approaches $n = 16$, wI_m approaches 1 since all teams fall within the “top 16” and ordering becomes deterministic. The convergence patterns across structures reveal their relative effectiveness at different ranking granularities.

Most structures exhibit monotonic increases in wI_m as K increases, reflecting the accumulating information about progressively larger subsets of the ranking. However, the group stage format (orange) displays a distinctive non-monotonic pattern characterized by sharp discontinuities at $K = 12$ and $K = 16$. These irregularities arise from the pot-based seeding structure: each group contains exactly one team from each pot (ranks 1-4, 5-8, 9-12, 13-16). This design creates clustering effects where teams finishing in specific group positions span multiple pots.

The sharp increases observed from $K = 11$ to $K = 12$ and $K = 15$ to $K = 16$ reflect the resolution of pot-based uncertainties as additional teams are included in the top- K tier. The plateaus between these transitions occur because teams eliminated in the same round receive randomly assigned ranks within their possible range. For instance, the four teams finishing last in their respective groups receive ranks 13 through 16, producing no additional

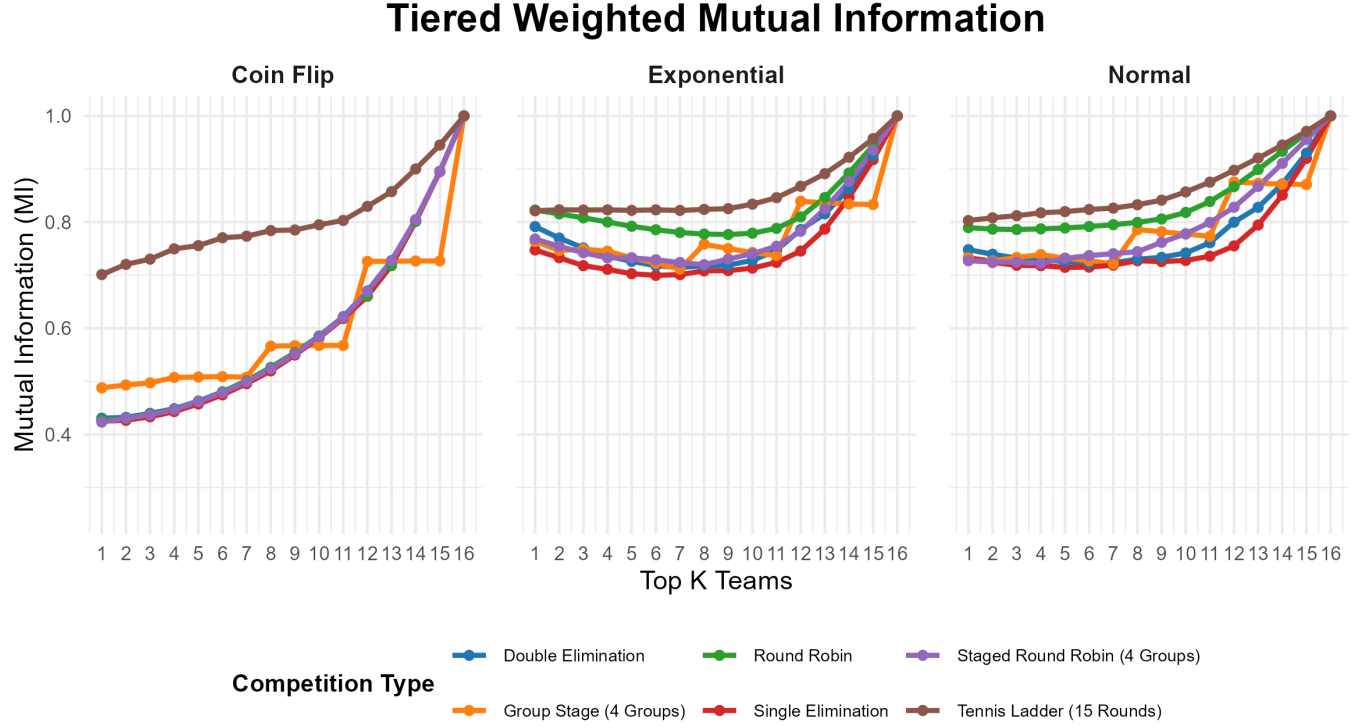


Figure 2: Tiered weighted mutual information of various competition structures as a function of the top K teams for three distributions of true strengths (Coin Flip, Exponential, Normal).

information until all four teams are included. While subdividing the final ranking into multiple tiers (e.g., knockout qualifiers, third-place finishers, last-place finishers) would eliminate these discontinuities, we maintained a unified ranking for cross-structure comparability.

6.3 Effects of Seeding Quality and Competition Variants

Beyond structure type, specific implementation choices significantly affect tournament informativeness. Figure 3 examines variants of single elimination and ladder structures under different seeding scenarios and series lengths.

6.3.1 Seeding Quality

Seeding quality exerts dramatically different effects depending on structure type. For the step ladder, seeding determines whether the structure achieves near-optimal performance or falls below random assignment. Under optimal seeding (top team plays winner of 2 vs 3 vs ... vs 16), the step ladder requires the best team to win only one match, the second-best two matches, and so forth, yielding exceptionally high wI_m values. However, under worst-case

seeding (1 vs 2, winner vs 3, etc.), the top team must win 15 consecutive matches to achieve the correct rank, while the second-best team will finish last if the top team wins the first match. This creates wI_m values below 0.25 for large K (worse than random assignment) because the structure systematically inverts the true ranking.

The strength distribution further modulates this effect. Under exponential distributions where top teams have higher win probabilities, badly-seeded step ladders show slightly elevated wI_m for small K , as the best team more reliably wins its 15 matches. Under more balanced distributions, even small K values produce poor results, as upsets become more probable throughout the extended gauntlet.

Traditional single elimination brackets prove remarkably robust to poor seeding when re-seeding is employed. A common example of a reseeded single elimination structure is the NFL playoff format. Figure 3 demonstrates minimal divergence between optimal and worst-case seeding under reseeding protocols, where remaining teams are rematched each round (highest remaining seed plays lowest, etc.). Reseeded formats actually slightly outperform standard brackets with optimal seeding, suggesting that dynamic seeding adjustments provide value even when initial seeding is correct. This finding indicates that when feasible, reseeded single elimination represents a more effective approach than fixed-bracket formats.

6.3.2 Series Length

Extending individual matchups from single games to best-of-series formats predictably increases wI_m across all structures and distributions. Longer series reduce outcome variance, allowing true strength differences to more reliably determine results. However, the magnitude of improvement varies by distribution: exponential distributions show larger gains from extended series than uniform distributions, as the longer format more reliably separates competitors with substantial skill gaps.

6.4 Round Robin Performance Under Extended Competition

When time and number of games is not a constraint, round robin is typically the most common approach to a structure. Examples of this are found most often in European soccer leagues, like the English Premier League.

Figure 4 examines round robin tournaments across multiple complete rounds, where each team plays every other team 1, 2, or 4 times. As the number of rounds increases, wI_m increases across all distributions, as repeated head-to-head matchups progressively reveal true strength rankings. Given infinite rounds, the metric would converge to 1 for any distribution with heterogeneous strengths.

The exponential distribution exhibits wider separation between curves as K increases, reflecting the varying difficulty of resolving rankings at different skill levels. Large strength differentials between top competitors are quickly resolved with high confidence, while similar bottom-tier competitors require more matchups to differentiate. This produces expanding gaps in wI_m as additional rounds disproportionately clarify the difficult-to-resolve lower ranks.

Conversely, the uniform distribution displays more consistent spacing between round-count curves and only modest variation around $K = 8$, reflecting the equal spacing of strength parameters. Since all pairwise differences are approximately equivalent, all rank pairs exhibit similar resolution difficulty, and increasing rounds improves rankings uniformly across the entire field. As round counts continue to increase beyond 4, these curves would be expected to flatten progressively.

6.5 Practical Implementation Considerations

While round robin with extensive rounds theoretically provides maximal information, real-world constraints on season length, athlete health, and scheduling feasibility necessitate more limited structures. To apply our metric to specific real-world competitions, we propose three potential approaches:

1. **Season-specific simulation:** Estimate strength parameters for multiple historical seasons, simulate tournament outcomes for each season independently, and aggregate results to compute an overall mutual information score
2. **Averaged metric approach:** Estimate strength parameters for multiple historical seasons, compute wI_m separately for each season’s simulations, and average the resulting metric
3. **Pooled strength distribution:** Estimate strength parameters across multiple seasons, aggregate them by rank position (e.g., average all championship teams’ parameters, all second-place teams’ parameters, etc.) to create a representative strength distribution, then simulate tournaments using this composite distribution.

Each approach offers different trade-offs between capturing seasonal variation and computational tractability, with the optimal choice depending on the specific application and available historical data.

6.6 Summary

Our weighted mutual information metric successfully quantifies tournament structure effectiveness in producing accurate rankings. No single structure universally dominates across

all scenarios: tennis ladders excel under ideal seeding, round robins provide maximal information given sufficient time, and reseeded single elimination offers a robust middle ground. The metric reveals that structure choice, seeding quality, strength distribution, and series length all interact to determine tournament informativeness. These results provide a quantitative foundation for evaluating existing tournament designs and optimizing structures given real-world constraints on time, resources, and competitive balance objectives.

7 Conclusion

This study introduced a weighted mutual information framework to evaluate the effectiveness of different competition formats in preserving the true underlying rankings of teams. We also introduce the concept of salient ranks, thereby extending and generalizing the concepts of efficacy and effectivity. By simulating outcomes of competition structures with varying seeding methods and applying an information-theoretic approach, we quantified how accurately each format conveys ranking information. Our findings confirm the intuitive advantage of round robin tournaments, which consistently outperformed single and double elimination formats in ranking accuracy, but also quantify the difference between these structures. Additionally, we demonstrated the degree to which poor seeding structures can significantly degrade performance of a competition structure, in some cases making competition outcomes even worse than simply randomly generating a set of ranks.

These results underscore the trade-off between tournament accuracy and practical constraints such as time, cost, and entertainment value. While round robin offers the highest fidelity to true rankings, it is often impractical for large tournaments, highlighting the need for hybrid or adaptive structures that balance accuracy with logistical feasibility.

Future work should extend this analysis to scenarios with more tournament structures, varying numbers of competitors, incorporate probabilistic strength models reflecting real-world uncertainty, and explore alternative weighting schemes for different competitive priorities (e.g., 4 salient ranks versus 8 salient ranking). Applying these methods to actual tournament data could further validate their usefulness for organizers aiming to design fair and informative competitions.

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Supplementary Material

All supplementary material available at <https://github.com/gjm112/tournaments>.

References

- Appleton, David R. 1995. “May the Best Man Win?” *Statistician* 44 (4): 529.
- Aziz, Haris, Serge Gaspers, Simon Mackenzie, Nicholas Mattei, Paul Stursberg, and Toby Walsh. 2017. “Fixing Balanced Knockout and Double Elimination Tournaments.” *Artificial Intelligence* 248: 1–14. <https://doi.org/10.1016/j.artint.2017.03.002>.
- Bradley, Ralph A., and Milton E. Terry. 1952. “Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons.” *Biometrika* 39 (3–4): 324–45. <https://doi.org/10.1093/biomet/39.3-4.324>.
- Carroll, Lewis. 1883. *Lawn Tennis Tournaments: The True Method of Assigning Prizes with a Proof of the Fallacy of the Present Method*. Online, transcription at schnark.github.io. <https://schnark.github.io/lewis-carroll/html/texts/lawn-tennis-tournaments-1883.html>.
- Cover, Thomas M., and Joy A. Thomas. 2006. *Elements of Information Theory*. 2nd ed. Wiley-Interscience.
- Csató, László. 2021. *Tournament Design: How Operations Research Can Improve Sports Rules*. 1st ed. Palgrave Macmillan.
- Devriesere, Karel, László Csató, and Dries R. Goossens. 2025. “Tournament Design: A Review from an Operational Research Perspective.” *European Journal of Operational Research* 324 (1): 1–21. <https://doi.org/10.1016/j.ejor.2024.10.044>.
- Devriesere, K., L. Csató, and D. Goossens. 2025. “Tournament Design: A Review from an Operational Research Perspective.” *European Journal of Operational Research* 324 (1): 1–21. <https://doi.org/https://doi.org/10.1016/j.ejor.2024.10.044>.
- Fagin, Ronald, Amnon Lotem, and Moni Naor. 2003. “Optimal Aggregation Algorithms for Middleware.” *Journal of Computer and System Sciences* 66 (4): 614–56. <https://www.biz.uiowa.edu/faculty/nstreet/Fagin03.pdf>.

- Glenn, W A. 1960. "A Comparison of the Effectiveness of Tournaments." *Biometrika* 47 (3-4): 253–62.
- Glickman, M. E., and A. C. Jones. 2025. "Models and Rating Systems for Head-to-Head Competition." *Annual Review of Statistics and Its Application* 12.
- Glickman, Mark E., and Shane T. Jensen. 2005. "Adaptive Paired Comparison Design." *Journal of Statistical Planning and Inference* 127 (1-2): 279–93. doi.org.
- Guiasu, Silviu. 1977. *Information Theory with Applications*. McGraw-Hill.
- Johnson, Sidney, and Rodney Fort. 2022. "Match Outcome Uncertainty and Sports Fan Demand: An Agnostic Review and the Standard Economic Theory of Ports Leagues." *Int. J. Emp. Econ.* 01 (02).
- Kendall, M. G. 1938. "A New Measure of Rank Correlation." *Biometrika* 30 (1/2): 81–93.
- Kendall, M. G., and B. B. Smith. 1940. "On the Method of Paired Comparisons." *Biometrika* 31 (3/4): 324–45.
- Kobayashi, Kei, Hideki Kawasaki, and Akimichi Takemura. 2006. "Parallel Matching for Ranking All Teams in a Tournament." *Advances in Applied Probability* 38 (3): 826–38.
- Krautmann, Anthony C, Young Hoon Lee, and Kevin Quinn. 2011. "Playoff Uncertainty and Pennant Races." *Journal of Sports Economics* 12 (5): 495–514.
- Lasek, Jan, and Marek Gagolewski. 2018. "The Efficacy of League Formats in Ranking Teams." *Stat. Modelling* 18 (5-6): 411–35.
- Mosteller, Frederick. 1952. "The World Series Competition." *Journal of the American Statistical Association* 47 (259): 355–80. <https://doi.org/10.2307/2281309>.
- Scarf, Philip A., and M. M. Yusof. 2011. "A Numerical Study of Tournament Structure and Seeding Policy for the Soccer World Cup Finals." *Journal of the Operational Research Society* 62 (10): 1874–84. <https://doi.org/10.1057/jors.2010.117>.
- Scheid, Francis. 1960. "A Tournament Problem." *American Mathematical Monthly* 67 (1): 39–41. <https://www.jstor.org/stable/2308923>.

- Shieh, Grace S. 1998. “A Weighted Kendall’s Tau Statistic.” *Stat. Probab. Lett.* 39 (1): 17–24.
- Slater, P. 1961. “Inconsistencies in a Schedule of Paired Comparisons.” *Biometrika* 48 (3/4): 303–12.
- Suksompong, Warut. 2016. “Scheduling Asynchronous Round-Robin Tournaments.” *Operations Research Letters* 44 (1): 96–100. <https://doi.org/10.1016/j.orl.2015.12.006>.
- Sziklai, Balázs R, Péter Biró, and László Csató. 2022. “The Efficacy of Tournament Designs.” *Comput. Oper. Res.* 144 (105821): 105821.

Single Elimination: Mutual Information

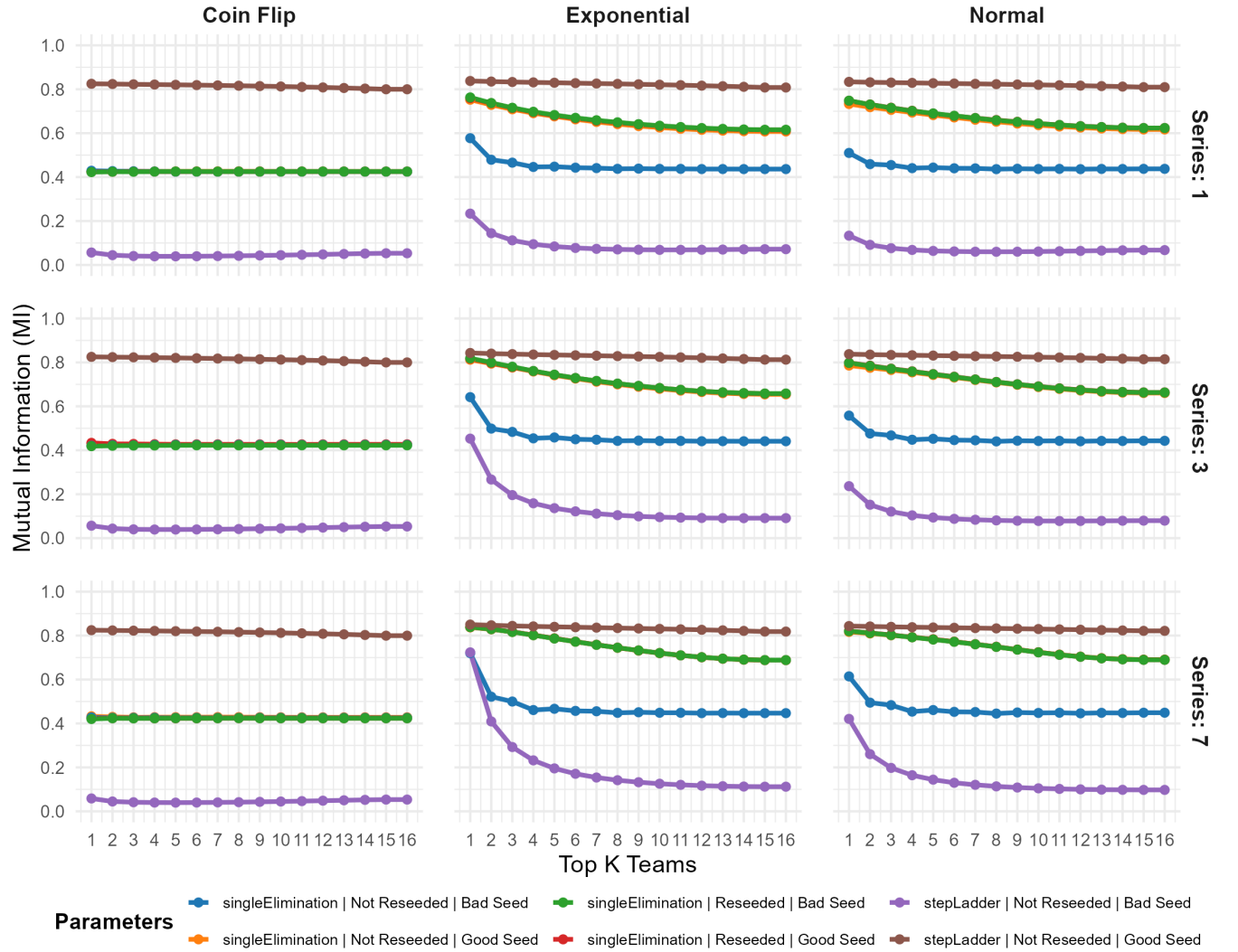


Figure 3: Mutual information of single elimination variants as a function of the top K teams for three distributions (Exponential, Normal, Coin Flip) and three series lengths (1, 3, 7).

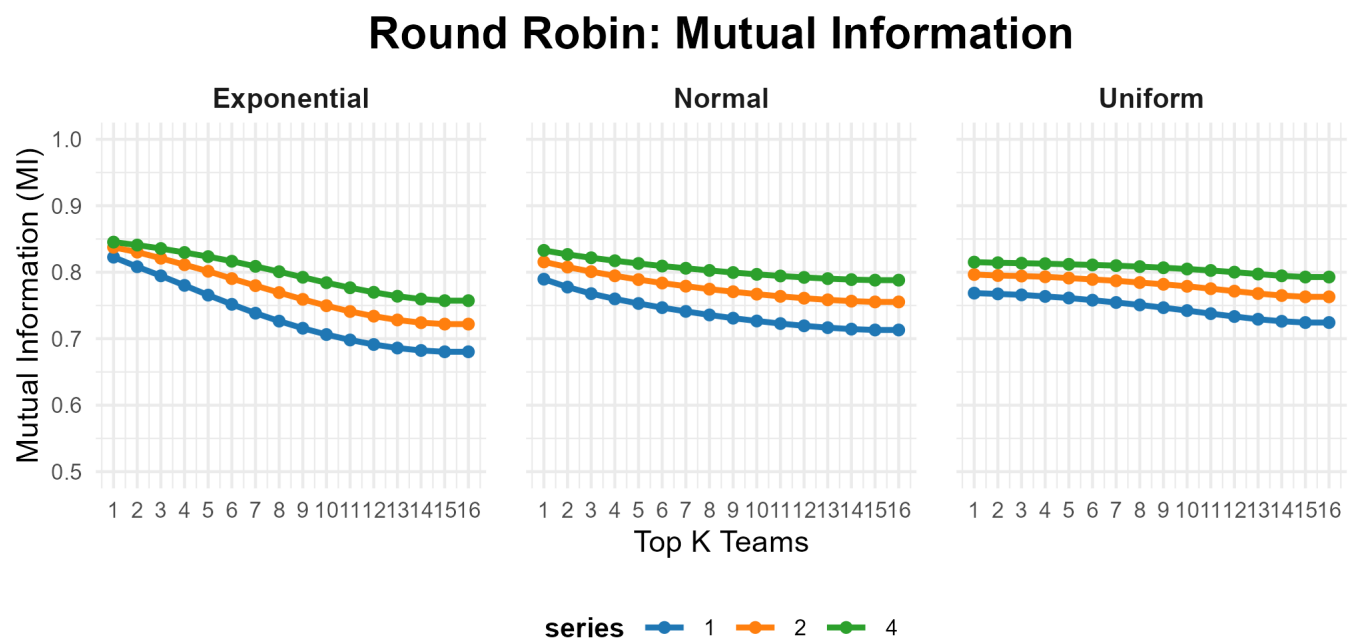


Figure 4: Mutual information of round robin structures as a function of the top K teams for three distributions (Exponential, Normal, Uniform) with varying numbers of complete rounds (1, 2, 4).